School of Computing
Queen's University

CISC 124

Winter 2026

Assignment 4

Deadline: Mar 11, 2026, 11:59 PM

"Rectangle" class

Design a class named `Rectangle` to represent a rectangle. The class contains:

- Two data fields named `width` and `height`.
- A constructor that creates a rectangle with the specified `width` and `height`. The default values are `1` and `2` for the `width` and `height`, respectively.
- A method named `getArea()` that returns the area of this rectangle.
- A method named `getPerimeter()` that returns the perimeter.

Draw the UML diagram for the class, and then implement the class. Write a test program that creates two `Rectangle` objects—one with width `4` and height `40` and the other with width `3.5` and height `35.7`. Display the width, height, area, and perimeter of each rectangle.

Assignment Submission

You are requested to submit two files as follows into OnQ Dropbox of Assignment 4:

1) The first file is a Word or PDF file containing:
    a. Write the following statement *" I 'your name' confirm that the submitted work in genuinely my work"*
    b. Analysis:
    (Describe the problem including input and output in your own words.)
    c. Design:
    (Draw an UML class diagram for the Rectangle class)
2) Coding of the program should have two parts:
    a. The implementation of the rectangle class in Java.
    b. Write a testing code that displays the width, height, area, and perimeter of each rectangle. Feel free to design your own format of output as long as you display the required data. (That is, display the width, height, area, and perimeter of Rectangle 1 and Rectangle 2)